

JKBOSE 2026

CLASS 11TH

BIOLOGICAL CLASSIFICATION

LECTURE-03

BOTANY

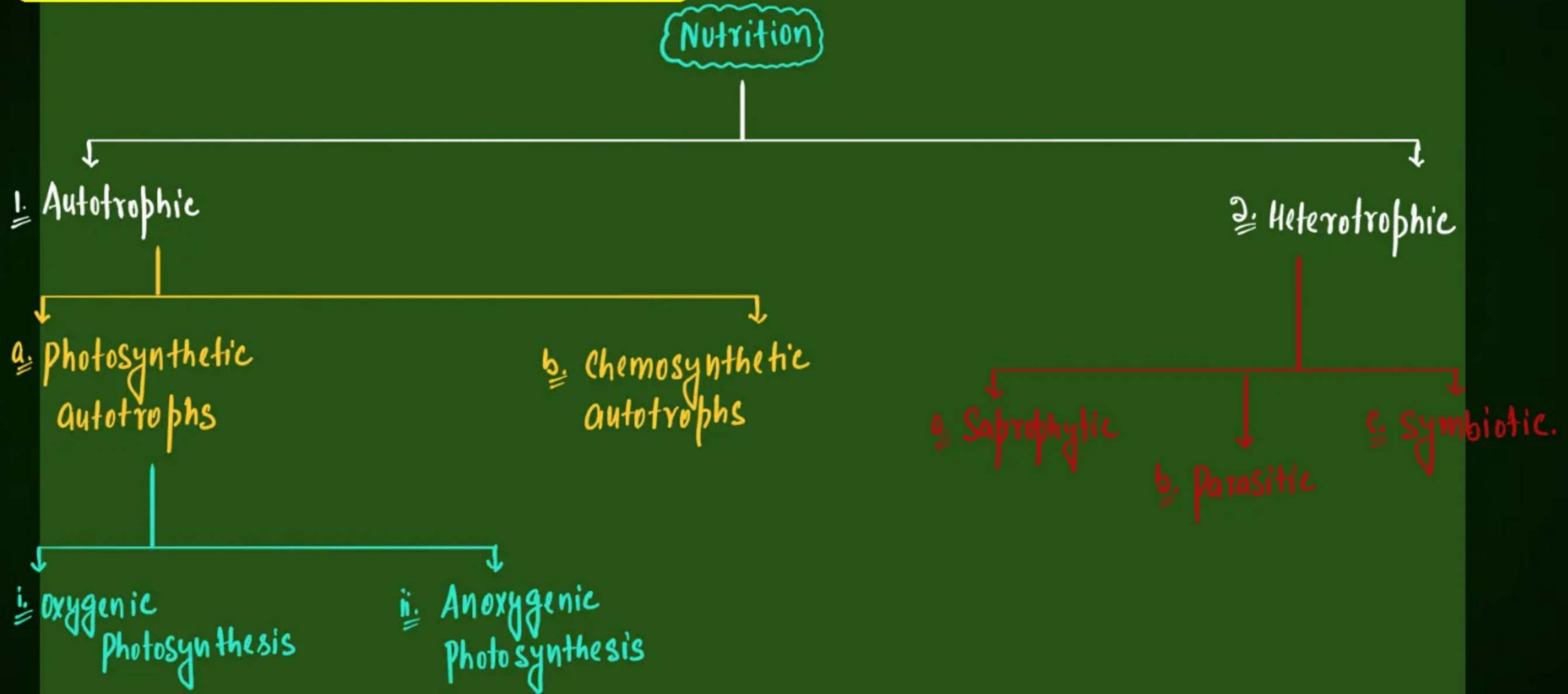
ARSHID BHAT SIR



Topics To Covered

1 Kingdom Monera (02) ✓

NUTRITION IN BACTERIA



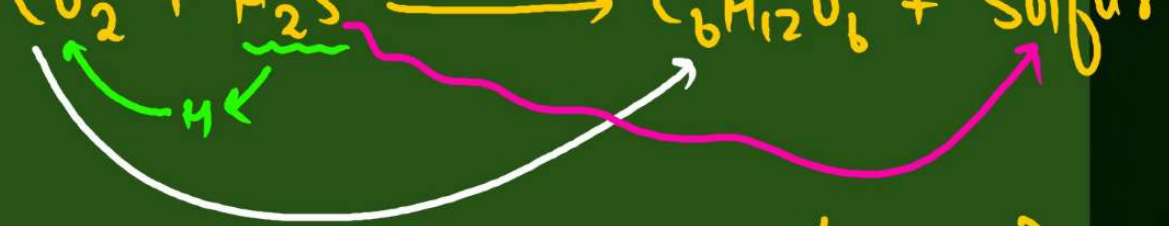
a. Photosynthetic autotrophs :-

- Source of energy is Sunlight.
- These type of bacteria possess photosynthetic pigments
- Depending on H-donor, it is of two types:

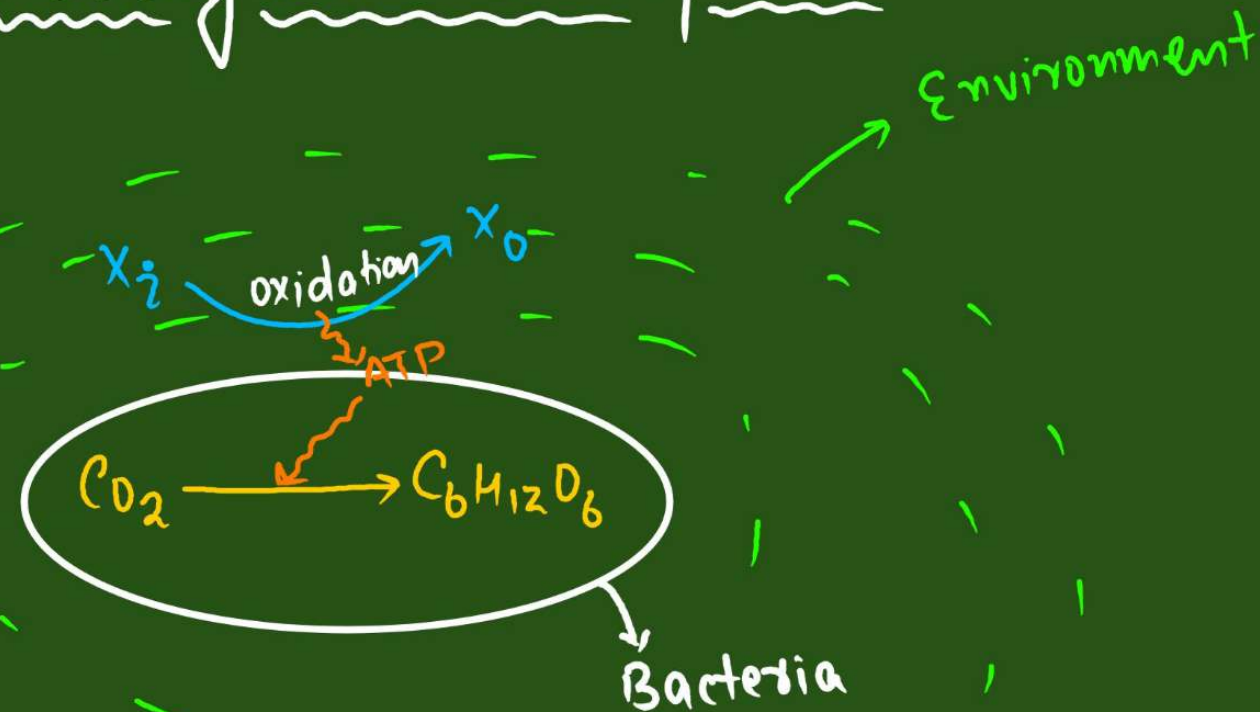
Oxygenic photoautotrophs:-

- $6\text{CO}_2 + 12\text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + \text{6O}_2$
By-product
- Here H-donor is H_2O .
- By-product is oxygen
- Ex. Cyanobacteria (Blue green algae).

An-oxygenic photoautotrophs:

- $\text{CO}_2 + \text{H}_2\text{S} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + \text{Sulfur}$

- H-donor is other than H_2O (ex. H_2S)
- By-product is other than O_2 (ex. Sulfur)
- Ex. Purple Sulfur Bacteria,
Green Sulfur bacteria...

b. Chemosynthetic Autotrophs :-



Source of energy = Chemical reactions
↓
occur outside bacterial cells (i.e., in their environment)

Ex. Nitrosomonas,
Nitrococcus,
Nitrobacter

Note:- These bacteria play an important role in recycling of $\underline{N}$, $\underline{P}$, $\underline{S}$, $\underline{Fe}$.

2. Heterotrophic mode of Nutrition :-

a. Saprophytic :-

≡ Source of food — dead and decaying organic matter

≡ Ex. Lactobacillus (helps in Curd formation).

b. Parasitic :-

• Source of food — other living organisms. (host organism).

• Majority of parasitic bacteria are pathogenic (i.e., disease causing).

• Bacterial diseases in animals — Cholera, Typhoid, pneumonia, tetanus etc.

• Bacterial diseases in plants — Crown gall disease, Citrus canker etc.

C. Symbiotic :-

• These bacteria lives symbiotically with other living organisms

• Ex. Rhizobium (Bacterium)

↓
lives symbiotically in roots of leguminous plants (ex. Pea)

- Rhizobium → fix atmospheric N_2 and provides it to plants.
- Plant (ex. Pea) → prepares food by photosynthesis and provides it to Rhizobium.

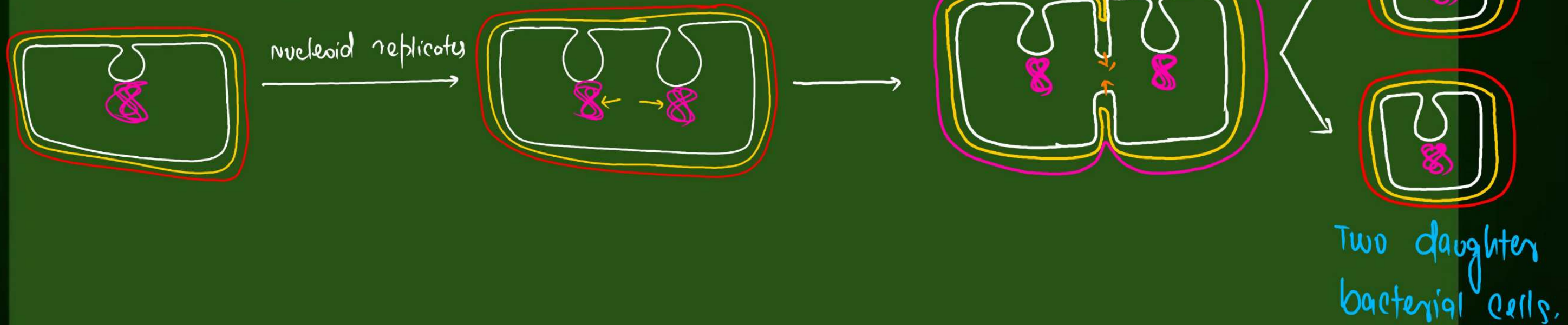
REPRODUCTION IN BACTERIA

→ Two types :

a. Asexual reproduction :-

- i. Binary fission;
- ii. Endospore formation.

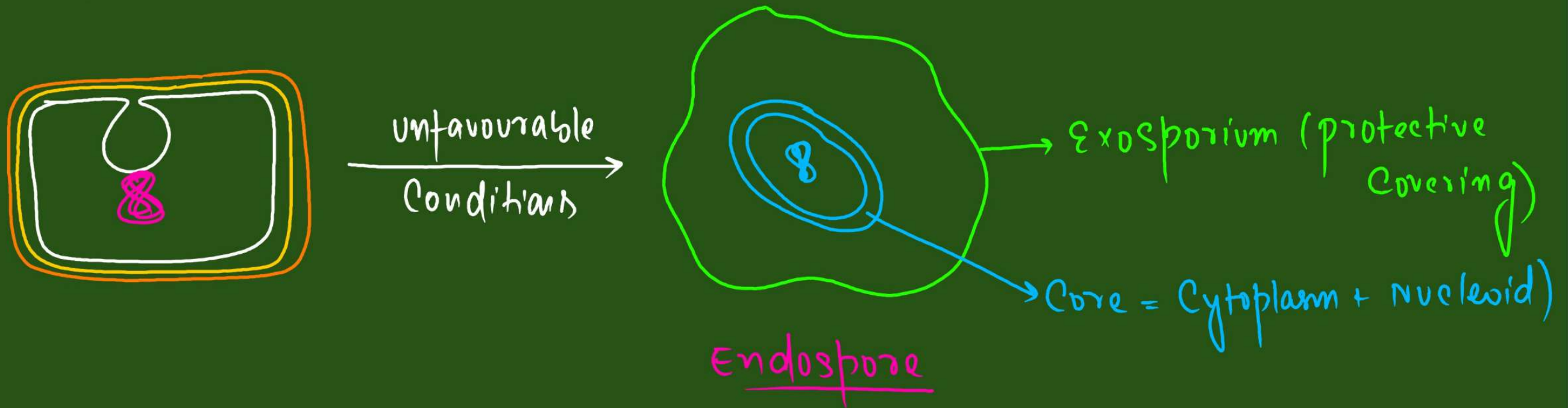
- ↓
- most common. ✓
 - occurs under favourable conditions. ✓



b. Sexual reproduction :-

ii. Endospore formation:- (Reproduction without multiplication)

- Occurs under unfavourable conditions. ✓
- Endospore — Easily tolerates $\pm 100^{\circ}\text{C}$ temp., toxic chemicals, high pH etc.



≡ Under favourable conditions — Exosporium gets degenerated.

≡ Endospore — liberated by air currents and upon germination forms a new bacterium.

Ex. Bacillus, Clostridium ...

B) Sexual reproduction:-

NEXT lecture ??

QUESTIONS

Oxygen is not produced during photosynthesis by

[2018]

- 1 *Nostoc*
- 2 Green sulphur bacteria
- 3 *Cycas*
- 4 *Chara*

QUESTIONS

U/W

BGA (blue green algae) are included in which of the following groups? [1996]

- 1 Bryophytes
- 2 Prokaryotes
- 3 Protista
- 4 Fungi

QUESTIONS

H/w

Chemosynthetic autotrophic bacteria:

- 1 Oxidize various inorganic substance such as ammonia and use the released energy for their ATP production
- 2 Play a great role in recycling of nutrients like nitrogen, phosphorus, iron and Sulphur
- 3 Both (1) and (2)
- 4 Can evolve oxygen during photosynthesis

QUESTIONS

Hw

The vast majority of bacteria are:

- 1 Symbiotic
- 2 Autotrophic
- 3 Heterotrophic
- 4 Nitrogen fixing

QUESTIONS

Hw

Cell wall of bacteria is made up of:

- 1 Cellulose
- 2 Polypeptide
- 3 Peptidoglycan
- 4 Pseudomurein

QUESTIONS

U/w

Most common method or reproduction in eubacteria is:

- 1 Binary fission
- 2 Multiple fission
- 3 Transduction
- 4 Transformation

Thank
You